

1. Introduction



Petroleum Refinery



Pharmaceutical plant



Fertilizer Plant



Food Processing Plant

Course Roadmap

01

Introduction

Chemical Engineering plant design, Overall design consideration, Practical considerations in design, engineering ethics in design

02

General Design Considerations:

Health and Safety hazards, Loss prevention, Environmental Protection, Plant Location, Plant Layout, Plant Operation and Control.

03

Process Design Development:

Development of design database, Process creation, Process design criteria, Process flow diagram (PFD), Equipment design specifications. Flow sheet synthesis and development: General procedure, Process information, Functions diagram, Flow sheet synthesis, Software use in process design.

04

Cost and asset accounting

General accounting procedure, Balance sheet and Income statements.

Course Roadmap

05

Analysis of Cost Estimation:

Cash flow for industrial operations, Factors affecting investment and production costs, Capital investments, Fixed capital and working capital, Estimation of capital investment, Cost indices, Estimation of total cost, Gross profit, Net profit and cash flow, Cost scaling factors, Net present value analysis

06

Interest and Insurance:

Interest, Simple interest, Compound interest, Nominal and effective interest rates, Continuous interest, Costs of capital, Time value of money, Annuity, Cash flow patterns, Income taxes, Present worth, Future worth, Taxes and Insurance.

07

Depreciation:

Depreciable investments, Methods for calculating Depreciation

08

Profitability Analysis:

General accounting procedure, Balance sheet and Income statements.

Course Roadmap

09

Optimum design and design strategy:

Defining the optimization problem, Selecting an objective function, Structural optimization, Parametric optimization, Variable screening and selection, Optimization Applications. Design Report.

How is a chemical plant designed?

Material and Energy balance



Designing the equipment



Arrangement



Connection



Measuring devices (flowmeter, thermocouples)

What is Process Design?

Process Design = Converting laboratory process into industrial scale.

Things to consider -

- Safety
- Economy
- Environmental concerns
- Operability
- Reliability

Plant Design

A plant-design project moves to completion through a series of stages such as is shown in the following:

1. Inception

The initial idea or need for a new plant/process is identified, often driven by market demand or a technological opportunity.

2. Preliminary evaluation of economics and market

Rough assessment of whether the product will be profitable, checking market demand, pricing, and approximate costs before committing resources.

3. Development of data necessary for final design

Collecting detailed technical data (lab/pilot plant results, physical/chemical properties, reaction kinetics) needed to design the process accurately.

4. Final economics evaluation

A more rigorous cost and profitability analysis using the detailed data, to decide whether to proceed with full-scale design.

Plant Design

5. Detailed engineering design

Preparing complete process flow diagrams, P&IDs, equipment specifications, and drawings for construction.

6. Procurement

Purchasing the equipment, materials, and machinery specified in the design.

7. Erection

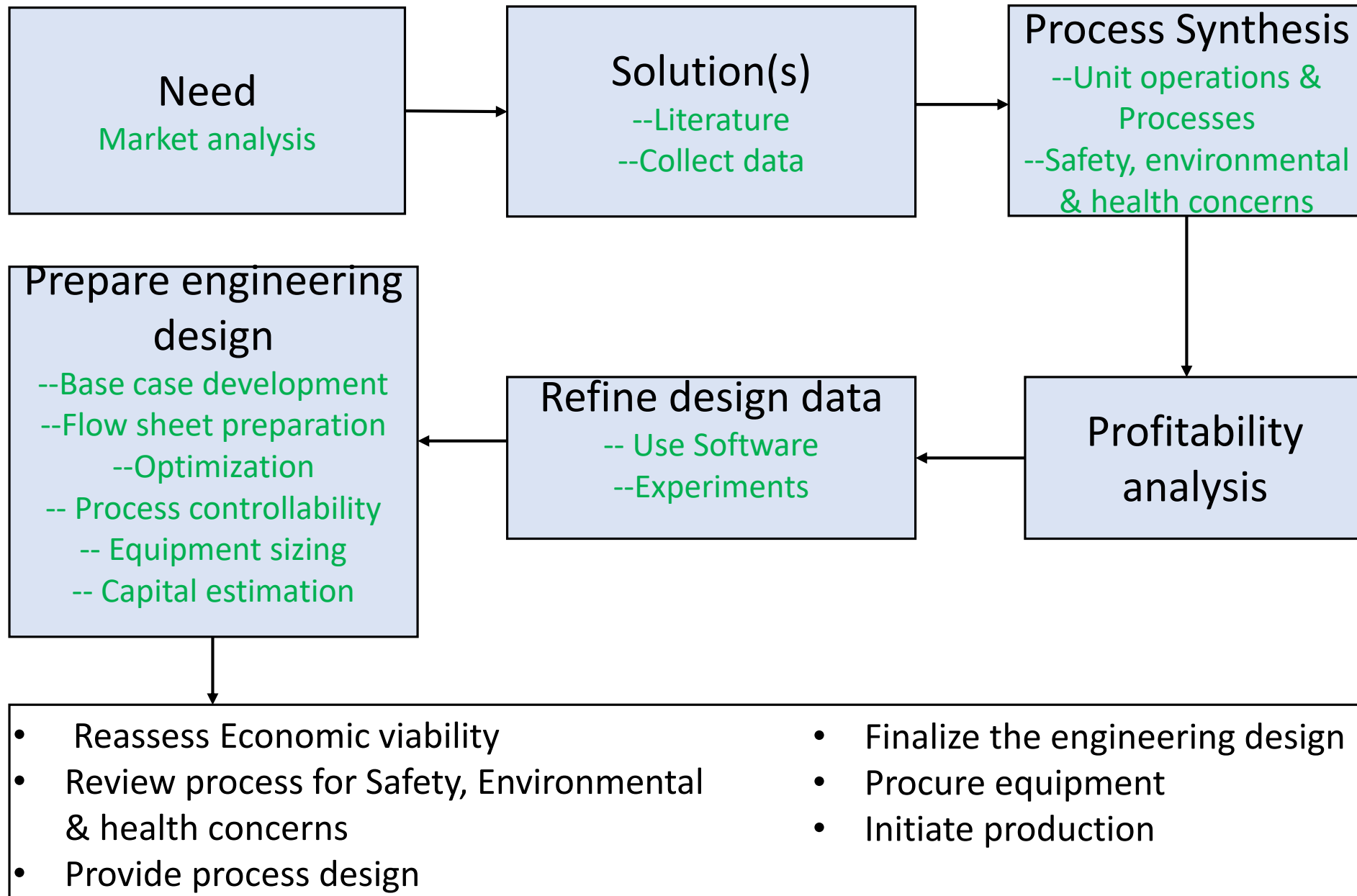
Physical construction and installation of equipment and piping at the plant site.

8. Startup and trial runs

Commissioning the plant, testing individual equipment and the overall system to iron out issues before full operation.

9. Production

The plant runs at full capacity, manufacturing the product for commercial sale.



Design Constraints

- Technical Constraints
- Safety
- Legal
- Raw material availability

Economic Constraints

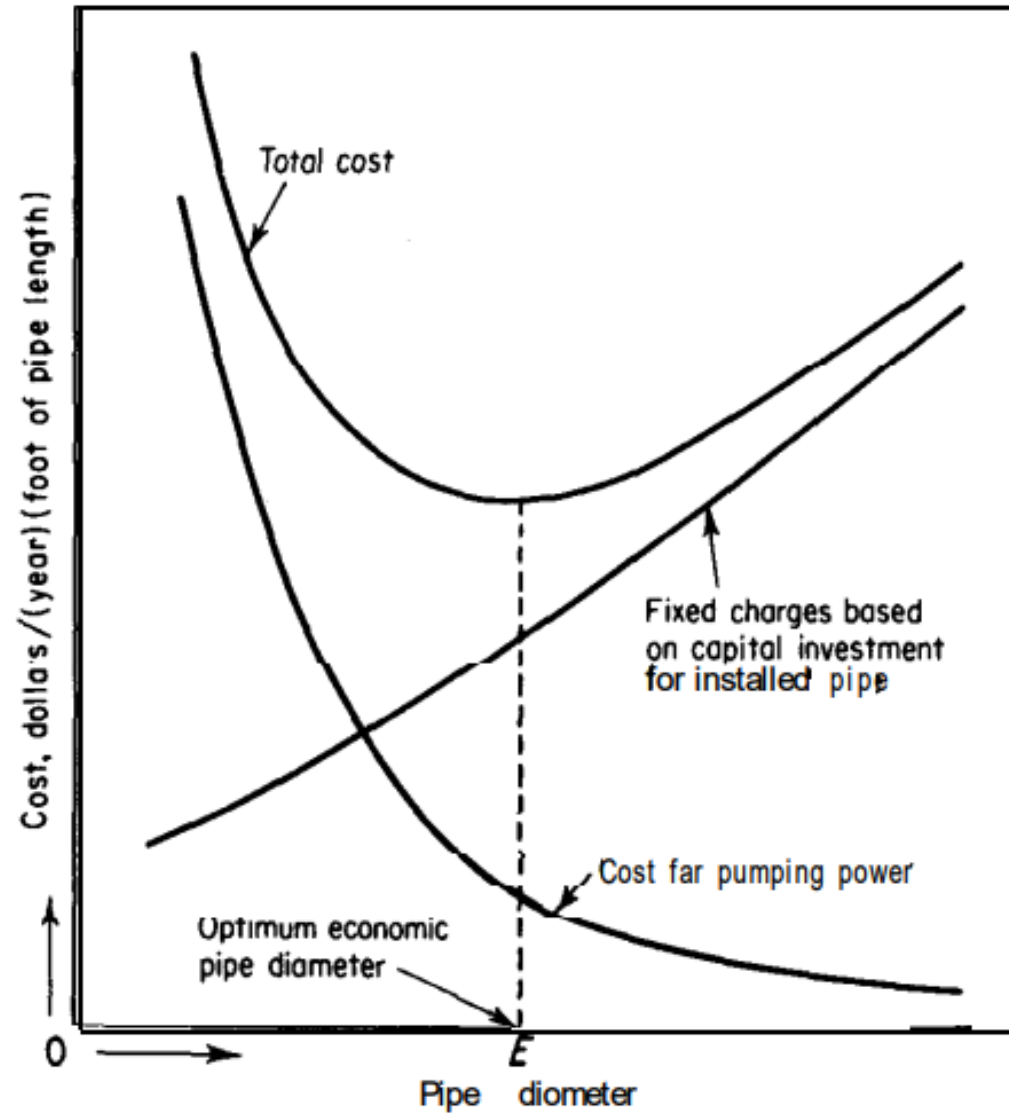
Environmental

Site and layout

Overall Design Consideration

- Plant location
- Plant layout
- Materials of construction
- Structural design
- Utilities
- Buildings and Storage
- Materials handling
- Safety and Waste disposal
- Federal, state, and local laws or codes, and
- Patents.

Optimum Economic Design



Feasibility Survey

Following is a list of items that should be considered in making a feasibility survey

- 1. Raw materials** (availability, quantity, quality, cost)
- 2. Thermodynamics and kinetics of chemical reactions involved** (equilibrium, yields, rates, optimum conditions)
- 3. Facilities and equipment available at present**
- 4. Facilities and equipment which must be purchased**
- 5. Estimation of production costs and total investment**
- 6. Profits** (probable and optimum, per pound of product and per year, return on investment)
- 7. Materials of construction**

8.Safety considerations

9.Markets (present and future supply and demand, present uses, new uses, present buying habits, price range for products and by-products, character, location, and number of possible customers)

10.Competition (overall production statistics, comparison of various manufacturing processes, product specifications of competitors)

11.Properties of products (chemical and physical properties, specifications, impurities, effects of storage)

12.Sales and sales service (method of selling and distributing, advertising required, technical services required)

13.Shipping restrictions and containers

14.Plant location

15.Patent situation and legal restrictions

Engineering Ethics

1. Highest priority – Safety, health and welfare of the public
2. Formally advise employees or clients about the consequences of their duties (if adversely affect their present and future health)
3. Accept the responsibilities and recognize the contribution of their work.
4. Always communicates facts honestly, accurately and without bias.
5. Act professionally
6. Treat all colleagues and co-workers fairly
7. Build professional reputation on the merit of services
8. Continue professional development and provide opportunities to others